better resources or evade threat. Courtship is a set of behavioral patterns to find a mate to reproduce. Most nocturnal animals remain underground or inactive during daytime. This is a modification of their feeding and activity pattern or habit or behaviour.

**Ethology** *is the scientific study of animal behaviour, under natural conditions.*

### c) Physiological adaptations

These are adaptations of organisms that help them to live and survive in their environment with unique niches. Example: Lions have sharp canines to hunt and tear meat and a digestive system suitable for digesting raw meat. The two most well-known physiological adaptations are **hibernation and aestivation**. These are two different types of inactivity where the metabolic rate slows down so much that the animal can survive without eating or drinking. Aquatic medium and terrestrial habitats have their own respective environmental conditions. Hence organisms have to evolve appropriate adaptations to select suitable habitats and niches.

#### Adaptations of aquatic animals

1. The pectoral fins and dorsal fins act as stabilizers or balancers and the caudal fin helps in changing the direction as a rudder.
2. Arrangement of body muscles in the form of bundles (myotomes) help in locomotion.
3. Stream lined structure helps in the swift movement of the animals in water.
4. Respiration by gills making use of gases dissolved in water.
5. Presence of air-bladders filled with air for buoyancy.
6. Presence of lateral-line system. They function as rheoreceptors which is helpful in echolocating objects in water.
7. Integuments rich in mucous glands are protected by scales.
8. Maintain water and ionic balance in its body with excretory structures.

#### Adaptations of terrestrial animals

1. Earthworms and land Planarians secrete a mucus coating to maintain a moist situation for burrowing, coiling, respiration, etc.,
2. Arthropods have an external covering over the respiratory surfaces and well-developed tracheal systems.
3. In vertebrate skin, there are many cellular layers besides the well protected respiratory surfaces that help in preventing loss of water.
4. Some animals obtain their water requirement from food as partial replacement of water lost through excretion.
5. Birds make nests and breed before the rainy season as there is availability of abundant food. But during drought birds rarely reproduce.
6. Camels are able to regulate water effectively for evaporative cooling through the skin and respiratory system and excrete highly concentrated urine, and can also withstand dehydration up to 25% of their body weight.

## 10.7 Populations




Population is defined as any group of organisms of the same species which can interbreed among themselves, and occupy a particular space and function as part of a biotic community. A population has various properties like population density, natality (birth rate), mortality (death rate), age distribution, biotic potential, dispersion and 'r', 'K' selected



XII Std Biology-Zoology Chapter-10 EM.indd 173 23/12/2022 13:07:07